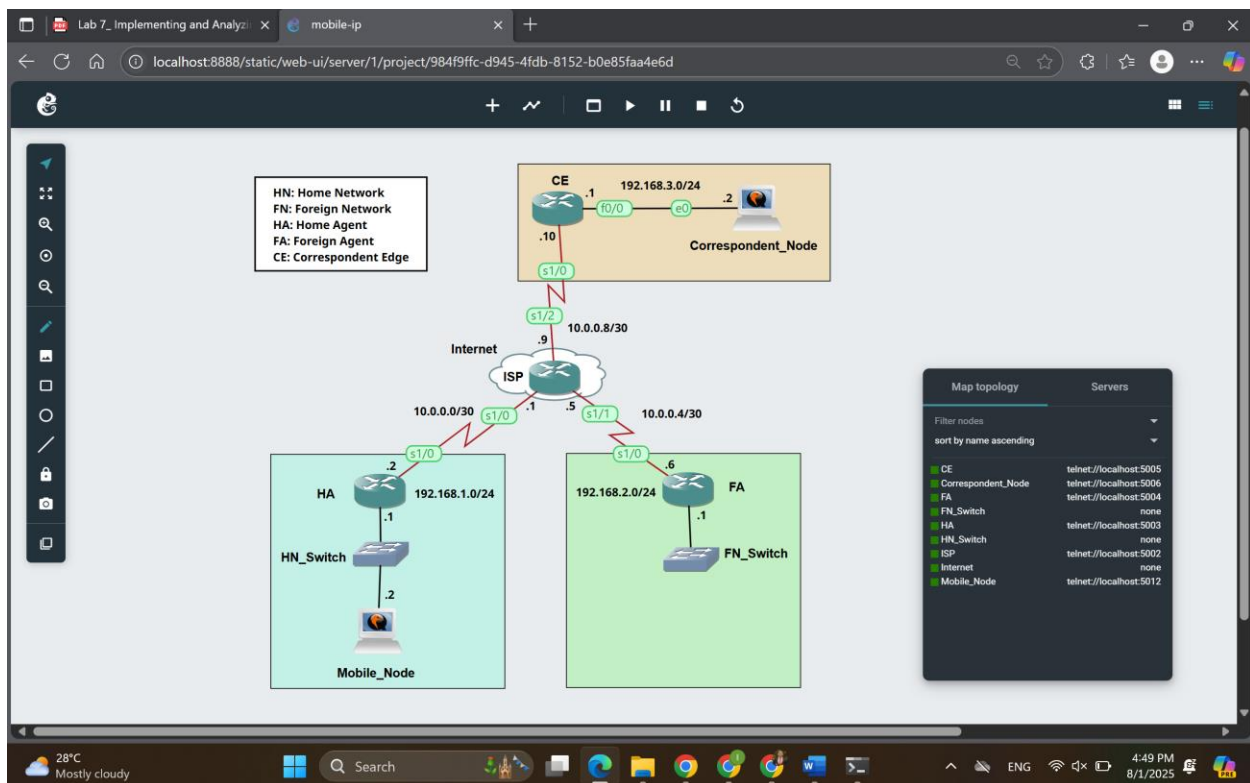


CO513: Advanced Computer Communication Networks - Lab 07

E/19/134, E/19/155, E/19/166, E/19/170,
E/19/205, E/19/210, E/19/309, E/19/349,
E/19/372, E/19/409, E/19/413, E/19/426,
E/19/ 432, E/19/443, E/19/446, E/19/452, E/19/455

25/07/2025

Task 1: Introduction and Setup



The above figure shows the initial topology setup for this lab. We used three client networks:

- Home Network 192.168.1.0/24
- Foreign Network 192.168.2.0/24
- Correspondent Network 192.168.3.0/24

all of which are connected to the core ISP network, represented by a single ISP router for this simulation.

The goal is to ensure that the Correspondent Node can connect to the Mobile Node, regardless of whether it is on the Home Network or Foreign Network with the same ip address.

For this lab simulations we have used

- Routers – Cisco C7200 Version 15.2(4) s5
- Switches – GNS3 L2 Ethernet Switch
- Hosts – Alpine Linux Release 3.20.0

Initial IP assignment in Routers

Home Agent

```
HA(config-if)#do show run | section int
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex full
interface Serial1/0
 ip address 10.0.0.2 255.255.255.252
 serial restart-delay 0
```

Foreign Agent

```
FA(config-if)#do show run | section int
interface FastEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 duplex full
interface Serial1/0
 ip address 10.0.0.6 255.255.255.252
 serial restart-delay 0
```

Correspondent Edge

```
CE(config-if)#do show run | section int
interface FastEthernet0/0
 ip address 192.168.3.1 255.255.255.0
 duplex full
interface Serial1/0
 ip address 10.0.0.10 255.255.255.252
 serial restart-delay 0
```

ISP

```
ISP(config)#do show run | section int
interface FastEthernet0/0
 no ip address
 shutdown
 duplex full
interface Serial1/0
 ip address 10.0.0.1 255.255.255.252
 serial restart-delay 0
interface Serial1/1
 ip address 10.0.0.5 255.255.255.252
 serial restart-delay 0
interface Serial1/2
 ip address 10.0.0.9 255.255.255.252
 serial restart-delay 0
```

For Mobile Node IP Assignment Setting DHCP

Inside Mobile Node (Alpine)

```
mobile:~# cat /etc/network/interfaces
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet dhcp
```

In Home Agent Router
configuring home network dhcp pool

```
HA#show run | section dhcp
ip dhcp excluded-address 192.168.1.1
ip dhcp pool HOME_POOL
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
 lease 0 2
HA#
```

In Foreign Agent Router
configuring foreign network dhcp pool

```
FA#show run | section dhcp
ip dhcp excluded-address 192.168.2.1
ip dhcp pool FOREIGN_POOL
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 8.8.8.8
 lease 0 2
FA#
```

Mobile Node received the dhcp ip configurations from the Home Agent as follows

```
mobile:~# ip addr
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
   link/ether 0c:d8:ad:2c:00:00 brd ff:ff:ff:ff:ff:ff
   inet 192.168.1.2/24 scope global eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::ed8:adff:fe2c:0/64 scope link
       valid_lft forever preferred_lft forever
```

For Correspondent Node IP Assignment Setting Static IP

Inside Correspondent Node (Alpine)

```
correspondent:~# cat /etc/network/interfaces
auto lo
iface lo inet loopback

auto eth0
iface eth0 inet static
 address 192.168.3.2
 netmask 255.255.255.0
 gateway 192.168.3.1
```

Task 2: Configure Basic IP Routing

Using OSPF Routing protocol to setup dynamic routing

In OSPF instead of specifying the subnet mask of a network it's complement (also known as wild card mask) is specified as shown in the following.

To achieve an ideal ISP and Customer Edge network topology as this one, typically requires a multi-area OSPF setup, which necessitates additional routing protocols like IGP support and can complicate all the rest of the configurations. However, for the simplicity of this simulation, the same default OSPF area, Area 0, was chosen for all the routers, ensuring that the main goal remain on mobile IP.

In Home Agent Router

```
HA(config)#do show run | section ospf
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
HA(config)#
```

In Foreign Agent Router

```
FA(config)#do show run | section ospf
router ospf 1
 network 10.0.0.4 0.0.0.3 area 0
 network 192.168.2.0 0.0.0.255 area 0
FA(config)#
```

In Correspondent Edge Router

```
CE(config)#do show run | section ospf
router ospf 1
 network 10.0.0.8 0.0.0.3 area 0
 network 192.168.3.0 0.0.0.255 area 0
CE(config)#
```

In ISP Router

```
ISP(config)#do show run | section ospf
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
ISP(config)#
```

Verification

Pinging from Mobile Node to
Correspondent Node

```
mobile:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2): 56 data bytes
64 bytes from 192.168.3.2: seq=0 ttl=61 time=45.212 ms
64 bytes from 192.168.3.2: seq=1 ttl=61 time=40.093 ms
64 bytes from 192.168.3.2: seq=2 ttl=61 time=35.167 ms
64 bytes from 192.168.3.2: seq=3 ttl=61 time=40.167 ms
64 bytes from 192.168.3.2: seq=4 ttl=61 time=35.127 ms
^C
--- 192.168.3.2 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 35.127/39.153/45.212 ms
mobile:~#
```

Pinging from Mobile Node to
Foreign Network

```
mobile:~# ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1): 56 data bytes
64 bytes from 192.168.2.1: seq=0 ttl=253 time=20.634 ms
64 bytes from 192.168.2.1: seq=1 ttl=253 time=25.496 ms
64 bytes from 192.168.2.1: seq=2 ttl=253 time=20.471 ms
64 bytes from 192.168.2.1: seq=3 ttl=253 time=25.352 ms
64 bytes from 192.168.2.1: seq=4 ttl=253 time=30.450 ms
^C
--- 192.168.2.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 20.471/24.480/30.450 ms
mobile:~#
```

This ensures OSPF routing is configured correctly and connectivity is successful between all the nodes. The following routing table which is taken from the Correspondent Edge router further clarifies that all the indirect networks in the topology are learned correctly via OSPF.

```
CE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O       10.0.0.0/30 [110/128] via 10.0.0.9, 00:02:52, Serial1/0
O       10.0.0.4/30 [110/128] via 10.0.0.9, 00:02:52, Serial1/0
C       10.0.0.8/30 is directly connected, Serial1/0
L       10.0.0.10/32 is directly connected, Serial1/0
O      192.168.1.0/24 [110/129] via 10.0.0.9, 00:02:52, Serial1/0
O      192.168.2.0/24 [110/129] via 10.0.0.9, 00:02:52, Serial1/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected, FastEthernet0/0
L      192.168.3.1/32 is directly connected, FastEthernet0/0
CE#
```

Task 3: Configure Mobile IP

On the Home Agent:

Tunnel Configuration

```
HA(config)#do show run | section Tunnel0
interface Tunnel0
 ip address 172.16.0.1 255.255.255.0
 tunnel source 10.0.0.2
 tunnel destination 10.0.0.6
HA(config)#
```

Home Address Binding (DHCP binding)

```
HA#show run | section MN
ip dhcp pool MN-STATIC
 host 192.168.1.2 255.255.255.0
 client-identifier 010c.d8ad.2c00.00
 default-router 192.168.1.1
 dns-server 8.8.8.8
```

Home Address Binding (route binding)

```
HA(config)#do show run | section ip route
ip route 192.168.1.2 255.255.255.255 Tunnel0
HA(config)#
```

On the Foreign Agent:

Tunnel (Forwarding Agent) Configuration

```
FA(config)#do show run | section Tunnel0
interface Tunnel0
 ip address 172.16.0.2 255.255.255.0
 tunnel source 10.0.0.6
 tunnel destination 10.0.0.2
FA(config)#
```

Care of Address Binding (DHCP binding)

```
FA#show run | section FN
ip dhcp pool FN-STATIC
 host 192.168.2.2 255.255.255.0
 client-identifier 010c.d8ad.2c00.00
 default-router 192.168.2.1
 dns-server 8.8.8.8
```

Care of Address Binding (route binding)

```
FA(config)#do show run | section ip route
ip route 192.168.1.2 255.255.255.255 192.168.2.2
FA(config)#
```

On the Mobile Node:

Since DHCP is enabled, the Mobile Node does not require any explicit configuration; IP assignment, agent discovery, and registration are handled automatically by the network.

Task 4: Test Mobility

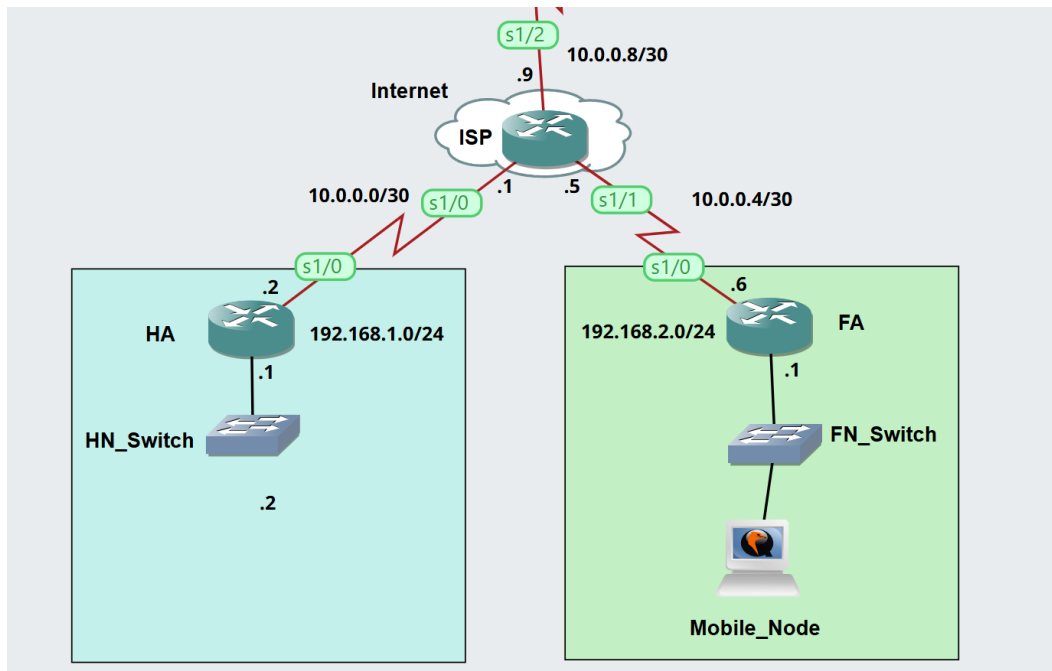
1. Before Movement

It can be seen from the output before the movement the Mobile Node can successfully ping Correspondent Node at 192.168.3.2.

```
mobile:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2): 56 data bytes
64 bytes from 192.168.3.2: seq=0 ttl=61 time=44.917 ms
64 bytes from 192.168.3.2: seq=1 ttl=61 time=39.790 ms
64 bytes from 192.168.3.2: seq=2 ttl=61 time=34.866 ms
64 bytes from 192.168.3.2: seq=3 ttl=61 time=39.910 ms
64 bytes from 192.168.3.2: seq=4 ttl=61 time=34.837 ms
64 bytes from 192.168.3.2: seq=5 ttl=61 time=39.939 ms
64 bytes from 192.168.3.2: seq=6 ttl=61 time=44.996 ms
64 bytes from 192.168.3.2: seq=7 ttl=61 time=50.082 ms
64 bytes from 192.168.3.2: seq=8 ttl=61 time=45.127 ms
64 bytes from 192.168.3.2: seq=9 ttl=61 time=50.109 ms
64 bytes from 192.168.3.2: seq=10 ttl=61 time=45.142 ms
64 bytes from 192.168.3.2: seq=11 ttl=61 time=50.278 ms
64 bytes from 192.168.3.2: seq=12 ttl=61 time=45.232 ms
64 bytes from 192.168.3.2: seq=13 ttl=61 time=40.138 ms
64 bytes from 192.168.3.2: seq=14 ttl=61 time=35.118 ms
64 bytes from 192.168.3.2: seq=15 ttl=61 time=40.120 ms
64 bytes from 192.168.3.2: seq=16 ttl=61 time=45.123 ms
64 bytes from 192.168.3.2: seq=17 ttl=61 time=50.019 ms
64 bytes from 192.168.3.2: seq=18 ttl=61 time=45.052 ms
64 bytes from 192.168.3.2: seq=19 ttl=61 time=40.014 ms
64 bytes from 192.168.3.2: seq=20 ttl=61 time=44.946 ms
64 bytes from 192.168.3.2: seq=21 ttl=61 time=49.976 ms
64 bytes from 192.168.3.2: seq=22 ttl=61 time=44.906 ms
```

2. Simulate Movement

The mobile node was detached from the home network as follows and connected to the foreign network.



The mobile node registering with the foreign agent happens automatically due to the dhcp configuration and the care-of-address was given successfully as follows.

```
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.2.2, server 192.168.2.1
udhcpc: lease of 192.168.2.2 obtained from 192.168.2.1, lease time 86400
```

The tunnel was enabled manually as follows.

```
HA(config)#interface Tunnel 0
HA(config-if)#no sh
HA(config-if)#
```

3. After Movement

It can be observed from the following traceroute output:

- The Home Agent (HA) established a tunnel to forward traffic from the MN's home address to its care-of address.
- Despite being away from its home network, the MN was able to maintain end-to-end IP connectivity with the **Correspondent Node (192.168.3.2)**.
- The multi-hop route includes both foreign and home networks, showing the encapsulated packets are decapsulated and delivered correctly via the tunnel.

```
mobile:~# traceroute 192.168.3.2
traceroute to 192.168.3.2 (192.168.3.2), 30 hops max, 46 byte packets
 1  192.168.2.1 (192.168.2.1)  7.515 ms  10.300 ms  9.860 ms
 2  10.0.0.5 (10.0.0.5)  19.822 ms  20.011 ms  19.879 ms
 3  10.0.0.10 (10.0.0.10)  29.869 ms  30.201 ms  29.857 ms
 4  192.168.3.2 (192.168.3.2)  39.987 ms  40.426 ms  39.948 ms
mobile:~#
```

The above hops are:

- **Hop 1:** FA's gateway which confirms MN is on the foreign network.
- **Hop 2-3:** Intermediate routers which belong to tunnel path.
- **Hop 4:** Final destination which is the Correspondent Host.

The following ping was carried out after the movement which further confirms stable connectivity.

```
mobile:~# ping 192.168.3.2
PING 192.168.3.2 (192.168.3.2): 56 data bytes
64 bytes from 192.168.3.2: seq=0 ttl=61 time=32.244 ms
64 bytes from 192.168.3.2: seq=1 ttl=61 time=37.433 ms
64 bytes from 192.168.3.2: seq=2 ttl=61 time=32.388 ms
64 bytes from 192.168.3.2: seq=3 ttl=61 time=37.447 ms
64 bytes from 192.168.3.2: seq=4 ttl=61 time=32.436 ms
64 bytes from 192.168.3.2: seq=5 ttl=61 time=37.650 ms
64 bytes from 192.168.3.2: seq=6 ttl=61 time=32.758 ms
64 bytes from 192.168.3.2: seq=7 ttl=61 time=37.906 ms
64 bytes from 192.168.3.2: seq=8 ttl=61 time=33.011 ms
```

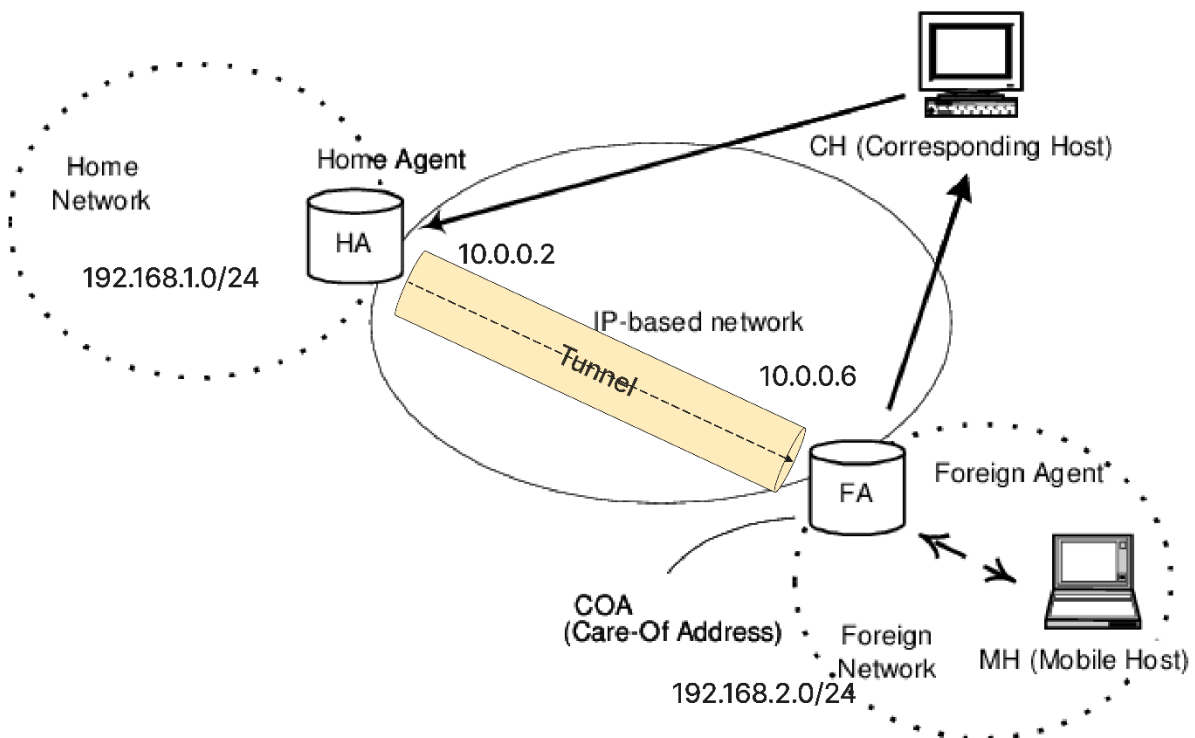

Task 5: Report and Reflection

1. What happens to the IP address of the MN after moving?

After moving to the foreign network, the Mobile Node (MN) retains its permanent home IP address, but it also temporarily obtains a Care-of Address (CoA) via the Foreign Agent (FA). Communication continues using the home address, while data is tunneled to the CoA.

2. How does the tunnel work?

The Home Agent (HA) intercepts packets sent to the MN's home address and encapsulates them (via GRE tunneling in our simulation). These packets are then forwarded to the MN's Care-of Address on the foreign network, where the FA decapsulates and delivers them to the MN. The following is an illustration of how our implemented tunnel works.



3. Advantages and Disadvantages of Mobile IP?

Advantages:

- Maintains persistent IP address and session continuity during mobility.
- Enables seamless roaming across networks.
- Transparent to applications (no reconfiguration needed).

Disadvantages:

- Triangular routing may cause increased latency (HA → FA → MN).
- Extra overhead due to tunneling.
- More complex configuration and potential security challenges.